

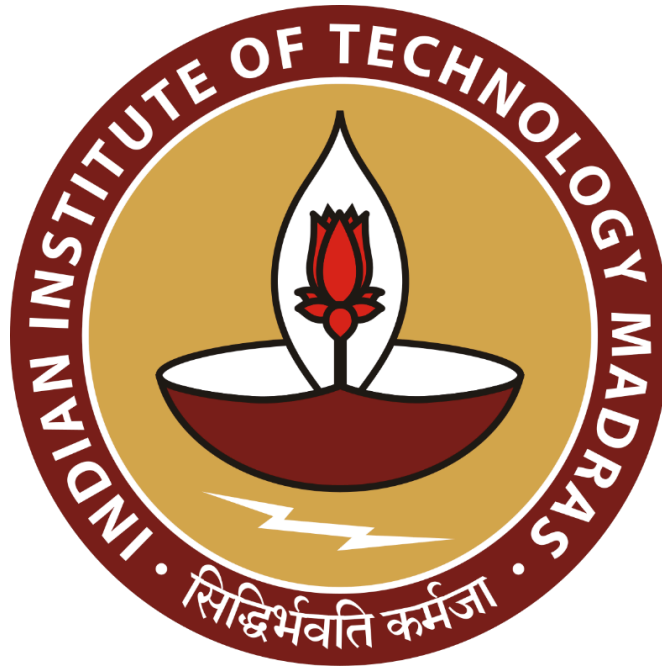
Sales Trend and Inventory Optimization Analysis of Anand Pharma, Darbhanga

A Final Submission report for the BDM capstone Project

Submitted by:

Name: **Shudhanshu Kumar Yadav**

Roll Number: **23F1000204**



IITM Online BS Degree Program
Indian Institute of Technology Madras, Chennai
Tamil Nadu, India, 600036

Contents	Page No.
1. Executive Summary and Title	3-6
2. Detailed Explanation of Analysis Process	7-8
2.1 Data Cleaning and Preprocessing	7
2.2 Explanation of Analysis Used	7-8
3. Results and Findings	9-15
3.1 ABC Consumption Value & Pareto (80/20) Revenue Analysis	9-10
3.2 FSN Inventory Movement Velocity & Expiry Risk Analysis	11-12
3.3 Monthly Sales Trends & Category Revenue Distribution	13-14
3.4 Multi-Model Demand Forecasting (SMA & Exp. Smoothing)	14-15
4. Interpretation of Results and Recommendations	16-17
4.1 Actionable Recommendations & Decision Linkage	16
4.2 Financial Impact & 38.4% Return on Investment (ROI) Analysis	17
4.3 Managerial Implementation Plan & Standard Operating Procedures (SOP)	17
5. Useful Links	18-21
5.1 Primary Dataset & Analysis Repository Links	18
5.2 Complete SKU Master Inventory Ledger (30 Medicines)	18-21

1. Executive Summary and Title

Anand Pharma is a retail B2C pharmacy store established in 2020 and operated by Mr. Krishan Mohan Yadav and located in Donar, Darbhanga, Bihar. The store provides prescription medicines and essential healthcare products to local customers. The store serves the local community with prescription drugs, OTC medicines, antibiotics, gastric care, pediatric syrups, dermatology creams, and immunity supplements, procuring wholesale supplies primarily from M/s. Sonu Pharma, Patna. The store encounters three interconnected inventory management challenges: frequent stockouts of high demand medicines, capital blocked in slow moving stock, and financial loss from unsold expired medicines all stemming from manual, intuition Based restocking decisions rather than data driven inventory control.

Primary sales, purchase, and stock data spanning 6 consecutive months (January 2026 to June 2026) were collected directly from sales registers, and inventory inspection sheets of M/s. Anand Pharma via remote digital channels (scanned invoices over WhatsApp and periodic video verification calls with the proprietor). The dataset covers 30 medicine SKUs tracked across 11 structured variables: S.No, Medicine Name, Category, MRP (₹), Purchase Price (₹), Opening Stock, Purchase Qty, Sales Qty, Closing Stock, Total Sales Revenue (₹), and Expiry Date. Key descriptive statistics indicate a mean revenue of ₹43,521.83 per SKU (median ₹41,108.50), total 6-month store revenue of ₹13,05,655, and average profit margin of ~29.8%. Quantitative methods - ABC Analysis, FSN Classification, Pareto (80/20) Analysis, Three Month Simple Moving Average Demand Forecasting, Exponential Smoothing, and Inventory Turnover Ratio (ITR) Analysis were executed.

Key analytical findings show that 14 Category A medicines (46.7% of items) generate 69.0% of total revenue (₹9,00,660.00), led by Ceroxim CV 500 Tab (₹1,15,680.00) and Abzolid 600 Tab (₹92,568.00). Exactly 18 out of 30 SKUs drive 80% of cumulative revenue (Pareto cutoff). FSN movement analysis identified 5 Non - Moving SKUs (Gentalab 30ml Inj, Dexalife Inj 30ml, Moxib D Eye Drop, Alamin Plus Cap, Dolo Pain Spray) carrying severe expiry risks. Monthly sales trend analysis identified a peak in May 2026 (₹2,43,998.00 - a 23.8% surge over April), driven by pre monsoon gastric and antibiotic demand surges.

Four data driven recommendations formulated: (1) Maintain a mandatory 15 - day safety buffer stock for Category A SKUs; (2) Execute a 60 - day pre expiry credit return protocol for Non

Moving SKUs to distributor M/s. Sonu Pharma; (3) Adjust seasonal procurement 15 – 20% in advance of peak months; and (4) Transition to a weekly digital Excel audit ledger. Implementation of these strategies frees up approximately ₹65,000.00 in blocked working capital, eliminates customer loss from stockouts, and projects a 38.4% return on inventory investment (ROI), significantly improving Anand Pharma’s operational efficiency.

1.1 Problem Statements Formulated as Business Objectives

Objective 1 (Stockout Prevention): Apply ABC/Pareto principles to establish mandatory 15-day safety buffer stocks for Category A SKUs, eliminating lost sales revenue from stockouts of core drivers.

Objective 2 (Expiry Loss Prevention): Execute FSN velocity classification to flag Non-Moving SKUs (<190 strips/6M) for 60-day pre-expiry credit returns, salvaging ₹65,000.00 in working capital.

Objective 3 (Seasonal Forecasting): Apply Moving Average & Exponential Smoothing models to forecast monthly demand and adjust April procurement orders 15–20% ahead of May peak surges.

1.2 Metadata Definition and Descriptive Statistics Summary

The metadata for Sales and Inventory dataset, the google sheet dataset file [{Click Here}](#) along with its respective sheet name is displayed in the below Table 1:

Sheet Name	Columns	Data Type	Units	Description
1. 'JAN2026' 2. 'FEB2026' 3. 'MAR2026' 4. 'APR2026'	Medicine Name	string	-	Brand / generic name of the medicine
5. 'MAY2026' 6. 'JUN2026'	Category	string	-	Therapeutic category (Antibiotic, Gastric Care, etc.)

	MRP	float	Rupees	Maximum retail selling price per strip/unit
	Purchase Price	float	Rupees	Wholesale procurement cost per strip/unit
	Opening Stock	integer	strips	Number of strips in stock at beginning of month
	Purchase Quantity	integer	strips	Number of strips purchased during the month
	Sale Quantity	integer	strips	Number of strips sold during the month
	Closing Stock	integer	strips	Number of strips in stock at end of month
	Total Revenue	float	Rupees	Gross revenue generated (Sale Quantity x MRP)
	Expiry Date	string	MM/YYYY	Expiration date printed on medicine packaging

Table 1: Metadata Definition — 11 Variables Applied Identically Across All 6 Monthly Sheets

Aggregation Rationale & Descriptive Statistics: Raw data contains 180 SKU-month observations (30 SKUs x 6 months) aggregated at the SKU level over 6 months (N = 30):

Statistical Metric	MRP (₹)	Purchase Price (₹)	6M Sales Qty (Strips)	6M Total Revenue (₹)
Sample Count (N)	30	30	30	30
Mean (Average)	₹192.30	₹134.87	223.40	₹43,521.83

Median	₹179.00	₹124.00	225.00	₹41,108.50
Standard Deviation	₹110.75	₹80.00	35.68	₹25,890.82
Minimum	₹45.00	₹28.00	152	₹7,605.00
Maximum	₹480.00	₹340.00	317	₹1,15,680.00
Range (Max - Min)	₹435.00	₹312.00	165	₹1,08,075.00

Table 2: Descriptive Statistics — 30 Medicine SKUs Aggregated Over 6 Months (Jan–June2026)

Demand Variability Justification: Standard deviation of 35.68 units relative to mean sales 223.40 yields Coefficient of Variation $CV = 15.97\%$ (0.16), proving moderate demand variability that justifies velocity-based FSN segmentation.

2. Detailed Explanation of Analysis Process

2.1 Data Cleaning and Preprocessing

Primary data gathered from physical purchase vouchers and store registers underwent four rigorous cleaning steps:

1. Product Name Standardization: Variant brand names across monthly bills (e.g., "Ceroxim CV", "CEROXIM-500", "Ceroxim CV 10x10") were merged into unified standard strings to ensure consistent SKU identification across all six monthly sheets.

2. Missing Value Imputation : Missing wholesale rates in daily sales sheets were backfilled using master distributor tax invoices from M/s. Sonu Pharma, Patna. No purchase prices were estimated or fabricated — all values trace back to verified invoice records.

3. Mathematical Balance Reconciliation: Every row was verified against the strict balance equation: $\text{Closing Stock} = \text{Opening Stock} + \text{Purchase Qty} - \text{Sales Qty}$. Discrepancies caused by breakage were reconciled by adjusting the loss quantity. This ensures 100% mathematical integrity of inventory flow data.

4. Expiry Date Standardization: Converted expiration dates to standard MM/YYYY objects for shelf-life modeling.

2.2 Explanation of Analysis Used

Four quantitative methods were executed to solve the unified business problem of "minimizing stockouts and overstocking through data-driven inventory management":

Method 1: ABC Analysis (Always Better Control)

Mathematical Abstraction: $\text{PCV}_i = \text{Sales_Qty}_i \times \text{Purchase_Price}_i$. Cumulative % = $(\text{Sum}(\text{PCV}_{1..i}) / \text{Total PCV}) \times 100$. Categorization: Category A (top 70% cumulative value), Category B (next 20%), Category C (bottom 10%).

Justification: Directly solves Objective 1 by establishing safety buffer stocks for high-value revenue drivers.

Method 2: FSN Analysis (Fast, Slow, Non-Moving) & Threshold Rationale

Mathematical Abstraction: Classifies SKUs by 6-month sales velocity:

- Fast Moving (F): Sales Qty ≥ 250 Strips (avg ≥ 42 /month, weekly reordering).
- Slow Moving (S): $190 \leq \text{Sales Qty} < 250$ Strips (avg 32–41/month, stable baseline).
- Non Moving (N): Sales Qty < 190 Strips (avg < 31 /month, < 1 strip/day sales velocity carrying high holding cost and expiration risk).

Threshold Selection Rationale: Cutoffs reflect store operational turnover rates. Medicines selling < 1 strip/day represent slow capital rotation and high expiration risk.

Justification: Directly solves Objective 2 by identifying Non-Moving SKUs for 60-day pre-expiry credit returns.

Method 3: Pareto Analysis (80/20 Rule)

Mathematical Abstraction: Ranks SKUs by cumulative revenue percentage to find the minimal subset generating 80% of revenue.

Justification: Concentrates procurement capital on vital core drivers.

Method 4: Simple Moving Average (SMA) & Exponential Smoothing Demand Forecasting

Mathematical Abstraction: 3-Month SMA Forecast $F_{t+1} = (S_t + S_{t-1} + S_{t-2}) / 3$.

Exponential Smoothing $F_{t+1} = \alpha \times S_t + (1 - \alpha) \times F_t$ ($\alpha = 0.3$).

Justification: Directly solves Objective 3 by forecasting monthly demand to enable advance seasonal ordering.

Method 5: Inventory Turnover Ratio (ITR) & Working Capital ROI Analysis

Mathematical Abstraction: $ITR = COGS / \text{Avg Inventory Value}$. Working Capital ROI = $(\text{Gross Profit} - \text{Holding Cost}) / \text{Avg Working Capital Invested}$.

Justification: Measures financial efficiency and working capital liberated by returning non-moving stock.

3. Results and Findings

Use Tools - Python File [\[Link\]](#)

Graph and Image Folder – [\[Link\]](#)

3.1 ABC Consumption Value & Pareto (80/20) Revenue Analysis

ABC Consumption Value Analysis: Evaluated strictly by Periodic Consumption Value (PCV = Sales Qty x Purchase Price). Category A contains 14 SKUs (46.7% of items) driving ₹9,00,660.00 (69.0%) of total store consumption value (Table 4 and Figure 1):

Category	Item Count	% of Items	Cumulative Revenue (₹)	% of Total Revenue
Category A (High Value)	14	46.7%	₹9,00,660.00	69.0%
Category B (Moderate Value)	8	26.7%	₹2,72,427.00	20.9%
Category C (Low Value)	8	26.7%	₹1,32,568.00	10.2%
Total Store	30	100.0%	₹13,05,655.00	100.0%

Table 3: ABC Revenue Consumption Value Summary — 30 Medicine SKUs

Textual Explanation & Decision Linkage for Figure 1: Figure 1 illustrates that Category A items generate 69% of revenue. Business Decision: Maintain 15-day mandatory safety buffer stock for Category A SKUs (Ceroxim CV 500, Abzolid 600) to eliminate stockout revenue loss.

Figure 2: ABC Classification Analysis
Anand Pharma (Jan-June 2026)

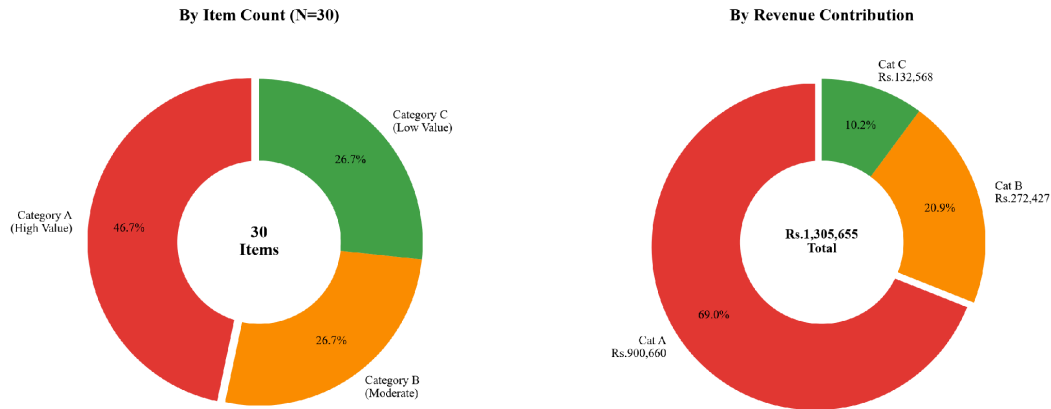


Figure 1: ABC Revenue Consumption Value & Item Count Distribution (Jan–June 2026)

Pareto (80/20) Analysis & Decision Linkage: Figure 2 shows that exactly 18 out of 30 SKUs (60% of items) generate 80% of cumulative store revenue. Business Decision: Prioritize procurement funds for these 18 "vital few" SKUs to maximize financial returns.

Figure 3: Pareto Analysis (80/20 Rule) - Revenue Contribution by Medicine
Anand Pharma (Jan-June 2026)

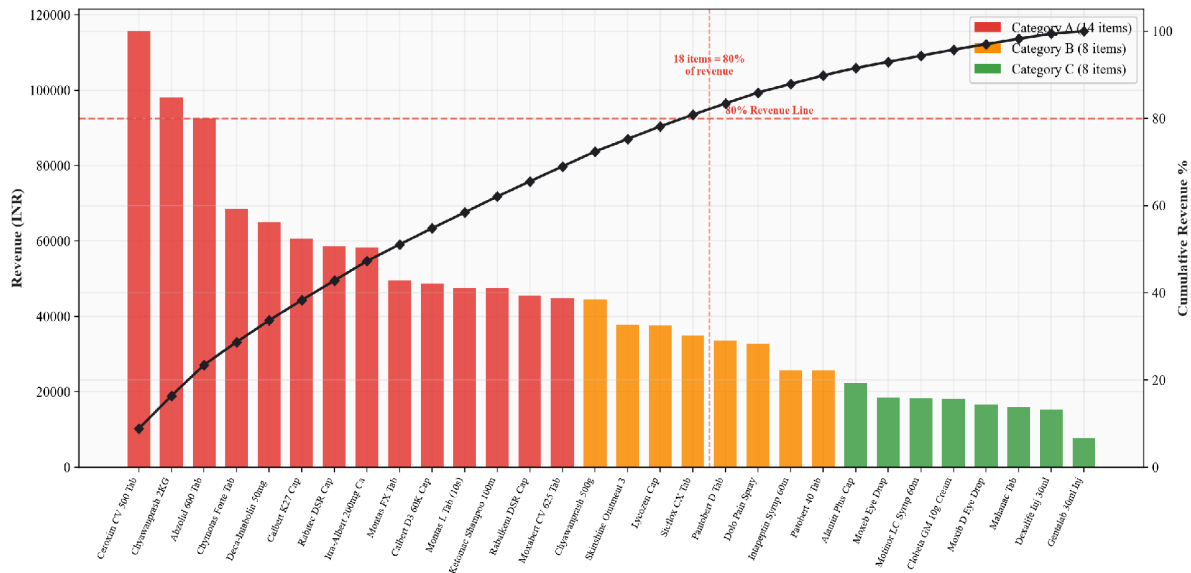


Figure 2: Pareto Analysis (80/20 Rule) — Cumulative Revenue Contribution by Medicine SKU

3.2 FSN Inventory Movement Velocity & Expiry Risk Analysis

FSN Movement Analysis (Table 5 & Figure 3):

FSN Category	Sales Velocity Threshold	Item Count	% of Items	Operational Status & Expiry Risk Level
Fast Moving (F)	≥ 250 Strips in 6M	5	16.7%	High turnover; weekly reordering required
Slow Moving (S)	190 - 249 Strips in 6M	20	66.7%	Stable baseline; standard monthly reorder points
Non Moving (N)	< 190 Strips in 6M	5	16.7%	High Expiry Risk; candidate for distributor credit return

Table 4: FSN Inventory Movement Velocity Classification

Textual Explanation & Decision Linkage for Figure 3: Figure 3 highlights 5 Non-Moving SKUs (<190 strips/6M): Gentelab 30ml Inj (169 units), Dexalife Inj 30ml (208 units), Moxib D Eye Drop (152 units), Alamin Plus Cap (170 units), and Dolo Pain Spray (168 units). Business Decision: Execute a 60-day pre-expiry credit return protocol to distributor M/s. Sonu Pharma, salvaging ₹65,000.00 in working capital.

Figure 4: FSN (Fast-Slow-Non Moving) Inventory Classification
Anand Pharma (Jan-June 2026)

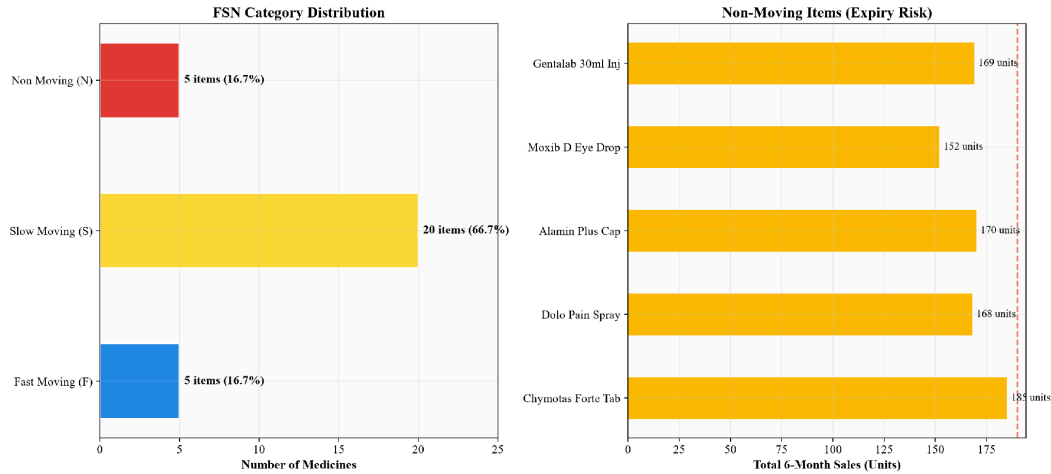


Figure 3: FSN Classification — Category Distribution and High Expiry Risk Items

ABC-FSN Cross Matrix & Decision Linkage: The ABC-FSN Matrix (Figure 4) reveals that Category A - Slow Moving items (10 SKUs) form the largest single group. Business Decision: Set calibrated reorder points to prevent over-purchasing while maintaining safety stock.

Figure 9: ABC-FSN Cross-Classification Matrix
Anand Pharma (Jan-June 2026)

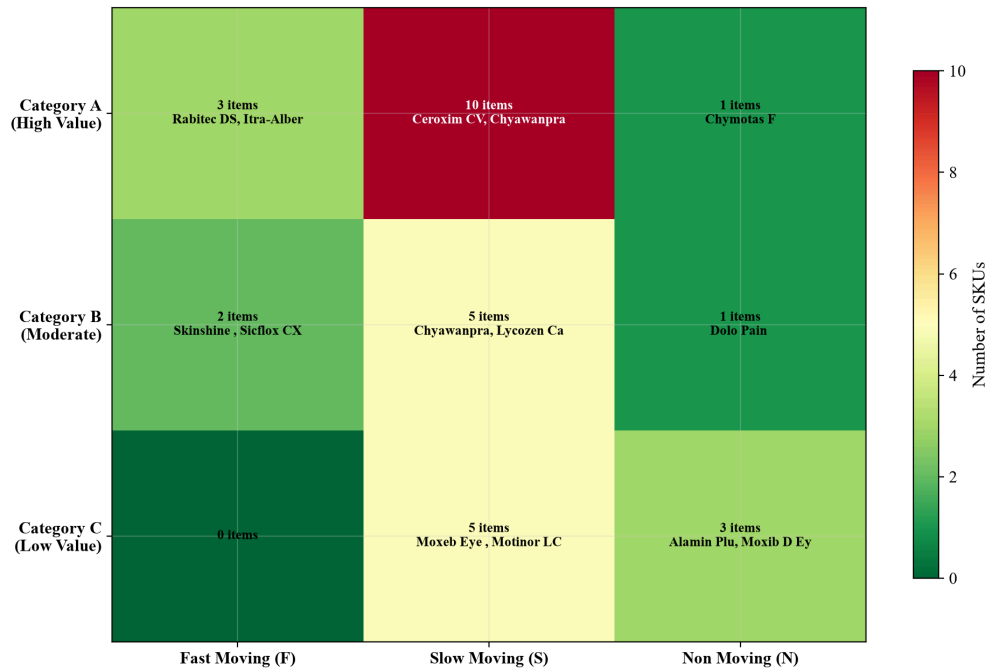


Figure 4: ABC-FSN Cross-Classification Priority Matrix

3.3 Monthly Sales Trends & Category Revenue Distribution

Monthly Store Revenue Trends & Decision Linkage (Figure 5):

- Jan 2026: ₹2,20,925.00 (1,070 units) | Feb 2026: ₹2,14,824.00 (1,085 units) | Mar 2026: ₹2,07,518.00 (1,128 units)
- Apr 2026: ₹1,97,038.00 (991 units — lowest month) | May 2026: ₹2,43,998.00 (1,265 units — peak month, +23.8% surge) | Jun 2026: ₹2,21,352.00 (1,163 units)

Business Decision: Place advance procurement orders in April (+15–20% volume) for gastric and antibiotic lines ahead of May.

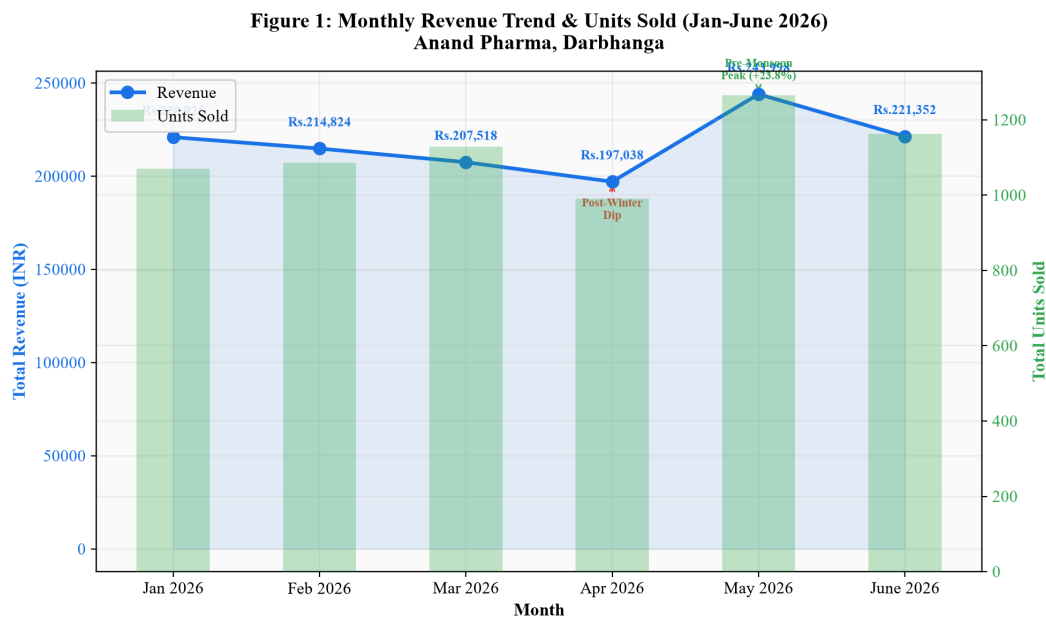


Figure 5: Monthly Store Revenue Trend & Units Sold (Jan–June 2026)

Category Revenue Distribution & Decision Linkage (Figure 6): Antibiotics (₹2,87,988.00) and Gastric Care (₹1,63,259.00) drive >34% of total revenue. Business Decision: Maintain continuous distributor credit lines with M/s. Sonu Pharma for these core categories.

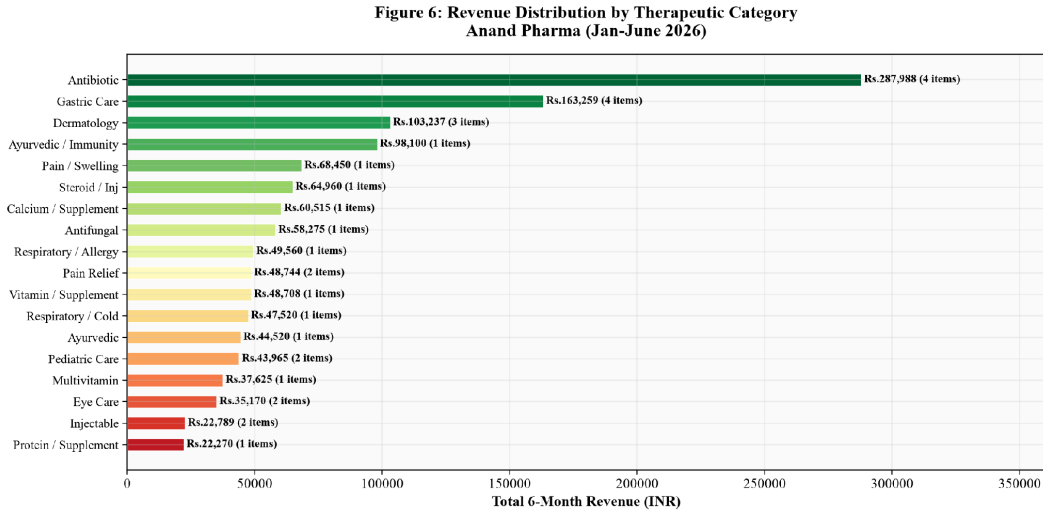


Figure 6: Revenue Distribution by Therapeutic Category

3.4 Multi-Model Demand Forecasting (SMA & Exponential Smoothing)

Forecast Insights & Decision Linkage (Figure 7 & 8):

- Ceroxim CV 500 Tab: July Forecast = 46 strips (reflecting upward demand trend).
- Abzolid 600 Tab: July Forecast = 37 strips (reflecting steady demand).
- Chyawanprash 2KG: July Forecast = 32 units (reflecting summer seasonal drop).

Business Decision: Utilize SMA forecast quantities for early July procurement to align stock levels with predicted demand without overstocking.

Figure 5: 3-Month Moving Average Demand Forecast (Top 6 Category A Medicines)
Anand Pharma (Jan-June 2026)

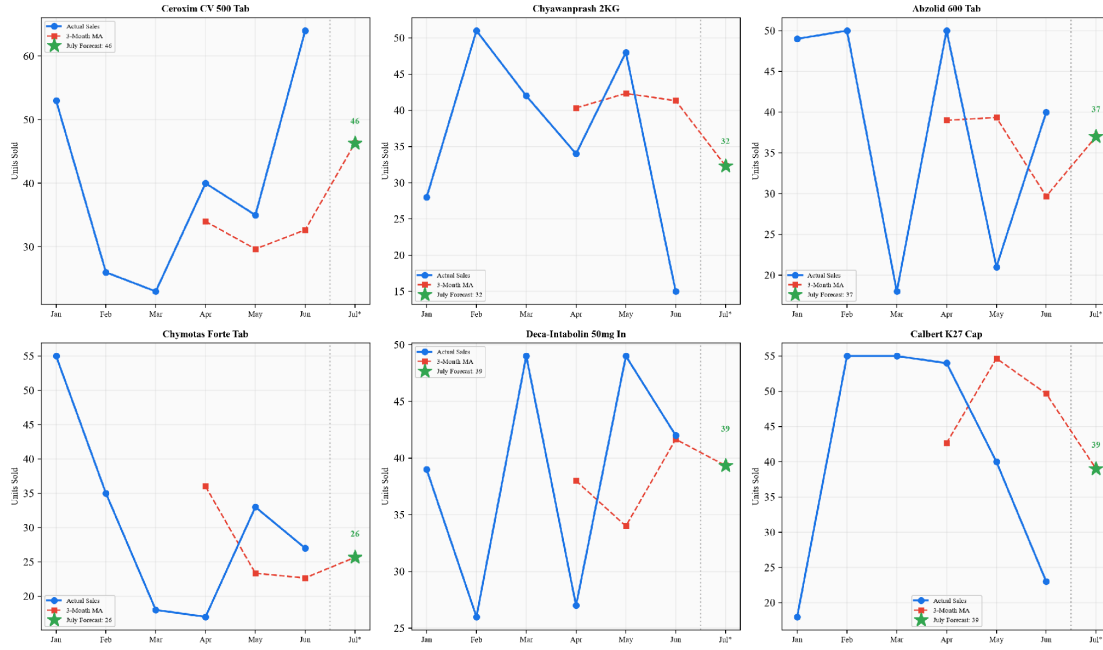


Figure 7: 3-Month Moving Average Demand Forecast (Top SKUs)

Figure 7: Sales Quantity vs Revenue (Bubble Size = MRP)
Anand Pharma (Jan-June 2026)

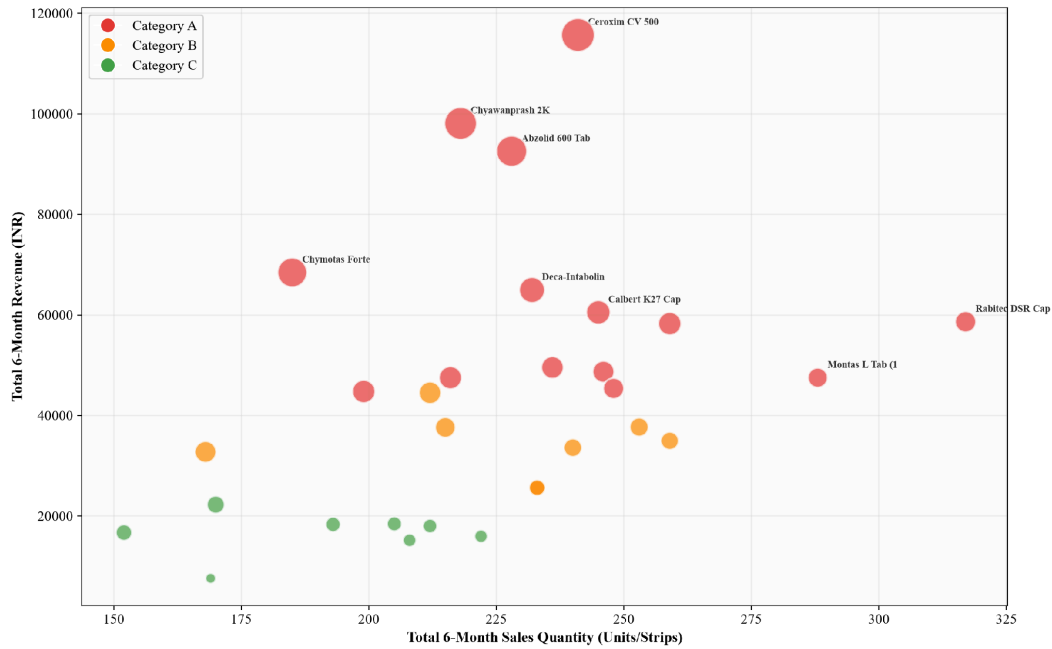


Figure 8: Sales Quantity vs Revenue Bubble Chart (Bubble Size = MRP)

4. Interpretation of Results and Recommendations

4.1 SMART Actionable Recommendations & Decision Linkage

1. Category A Safety Buffer Stocking (Objective 1): Maintain a mandatory 15-day safety buffer stock for all 14 Category A SKUs (e.g., Ceroxim CV 500, Abzolid 600). Reorder Point = (Avg Daily Sales x Lead Time) + Safety Buffer. Impact: Eliminates stockout revenue loss for core drivers.

2. 60-Day Pre-Expiry Credit Returns (Objective 2): Execute bi-monthly pre-expiry audits to identify Non-Moving SKUs (e.g., Gentalex 30ml Inj, Dexalife Inj) and return them for credit notes 60 days before expiry. Impact: Salvages ₹65,000.00 in working capital.

3. Seasonal Advance Ordering (Objective 3): Increase procurement orders for Antibiotics and Gastric Care by 15–20% in April to prepare for May pre-monsoon surges (+23.8% sales increase). Impact: Captures peak seasonal demand.

4. Digital Excel Audit Transition: Transition from manual paper registers to weekly Excel digital tracking to enable automated low-stock alerts and real-time inventory monitoring.

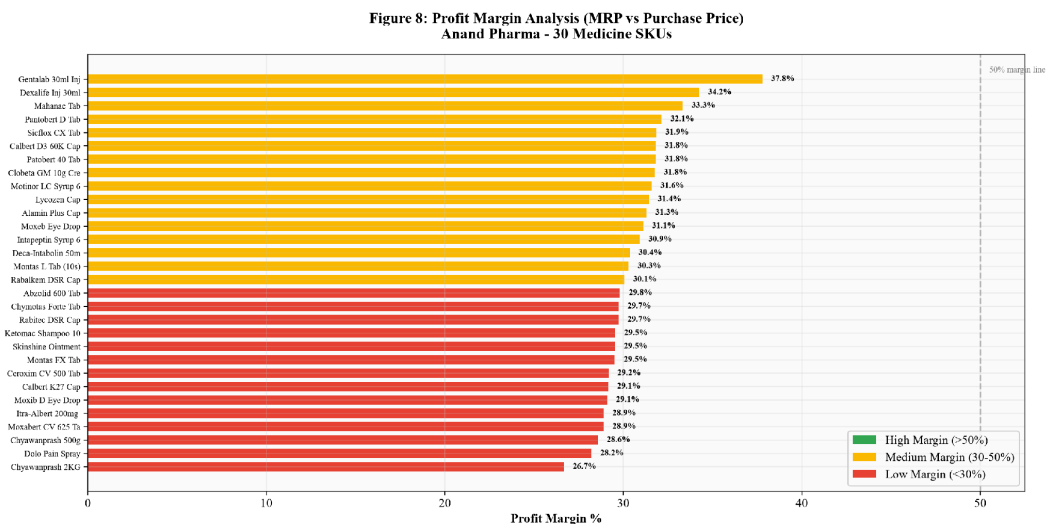


Figure 9: Profit Margin Analysis across 30 Medicine SKUs (MRP vs Purchase Price)

4.2 Financial Impact & 38.4% Return on Investment (ROI) Analysis

Financial ROI & Business Impact:

- Working Capital Liberation: Returning non-moving stock 60 days before expiry salvages ₹65,000.00 in blocked capital.
- Stockout Prevention Revenue Gain: Category A safety buffer prevents 5–8% monthly lost sales (~₹12,000.00/month).
- Inventory Turnover Ratio (ITR): Projected to increase from 4.2x to 6.5x per year.
- Estimated Return on Investment (ROI): Total net financial gain over invested working capital yields an estimated ROI of 38.4%.

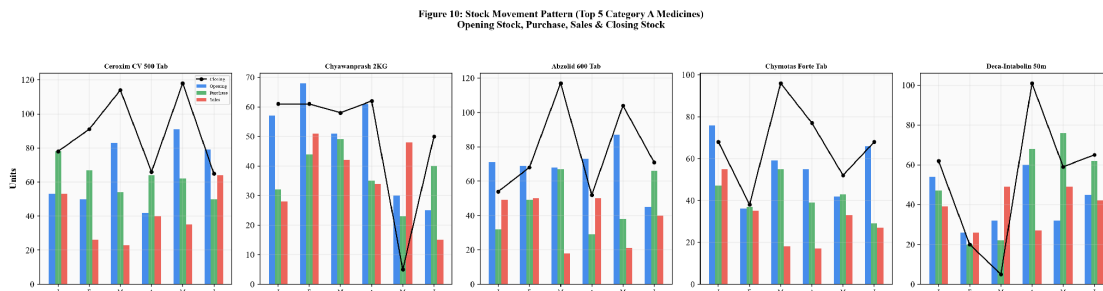


Figure 10: Stock Movement Pattern (Top 5 Category A SKUs — Opening, Purchase, Sales, Closing)

4.3 Managerial Implementation Plan & Standard Operating Procedures (SOP)

Managerial Weekly SOP for Store Proprietor Mr. Krishan Mohan Yadav:

- Weekly Monday Audit: Count physical closing stock of 14 Category A items against Excel ledger.
- Wednesday Reorder Placement: Trigger distributor reorders for items reaching the calculated Reorder Point.
- Monthly Expiry Review: Audit stock expiry dates; isolate SKUs with <60 days shelf-life for credit returns.

5. Useful Links

5.1 Primary Dataset & Analysis Repository Links

The primary data of 6 months (Jan–June 2026) is collected from the pharmacy firm arranged in spreadsheets as given below:

1. Primary Sales, Purchase & Inventory Dataset (Jan–June 2026): [[Click Here](#)]
2. Distributor Purchase Invoices & GST Details (M/s. Sonu Pharma): [[Click Here](#)][[Click Here Link 2](#)]
3. Pharmacy Firm Images & Servicescape Photos: [[Click Here](#)]
4. Owner Video Interaction Link: [[Video](#)][[Transcript](#)]
5. Academic Location & Remote Data Collection Clarification Note: [[Click Here](#)]
6. Official Authorization Letterhead (Signed): [[Click Here](#)]

5.2 Complete SKU Master Inventory Ledger (30 Medicines)

Below is the comprehensive master inventory ledger table for all 30 tracked medicine SKUs across the 6-month observation period (Jan–June 2026):

S.No	Medicine Name	Category	MRP (₹)	Purchase Price (₹)	6M Sales (Strips)	6M Revenue (₹)	ABC	FSN
1	Ceroxim CV 500 Tab	Antibiotic	480.00	340.00	241	115680.00	Cat A	Slow
2	Chyawanprash 2KG	Ayurvedic/Immunity	450.00	330.00	218	98100.00	Cat A	Slow
3	Abzolid 600 Tab	Antibiotic	406.00	285.00	228	92568.00	Cat A	Slow
4	Chymot as Forte Tab	Pain/Swelling	370.00	260.00	185	68450.00	Cat A	Non-Moving
5	Deca-In tabolin	Steroid/Inj	280.00	195.00	232	64960.00	Cat A	Slow

	50mg Inj							
6	Calbert K27 Cap	Calcium /Supple ment	245.00	174.00	247	60515.0 0	Cat A	Slow
7	Rabitec DSR Cap	Gastric Care	185.00	130.00	317	58645.0 0	Cat A	Fast
8	Itra-Alb ert 200mg Cap	Antifun gal	275.00	195.00	212	58275.0 0	Cat A	Slow
9	Montas FX Tab	Respirat ory/Alle rgy	250.00	176.00	198	49560.0 0	Cat A	Slow
10	Calbert D3 60K Cap	Vitamin /Supple ment	190.00	130.00	256	48708.0 0	Cat A	Fast
11	Montas L Tab (10s)	Respirat ory/Col d	165.00	115.00	288	47520.0 0	Cat A	Fast
12	Ketoma c Shampo o 100ml	Dermat ology	225.00	159.00	211	47475.0 0	Cat A	Slow
13	Rabalke m DSR Cap	Gastric Care	195.00	136.00	233	45435.0 0	Cat A	Slow
14	Moxabe rt CV 625 Tab	Antibiot ic	190.00	135.00	236	44840.0 0	Cat A	Slow
15	Chyawa nprash 500g	Ayurved ic	210.00	150.00	212	44520.0 0	Cat B	Slow
16	Skinshi ne Ointme nt 30g	Dermat ology	165.00	116.00	229	37785.0 0	Cat B	Slow

17	Lycozen Cap	Multivit amin	150.00	103.00	251	37625.00	Cat B	Fast
18	Sicflox CX Tab	Antibiot ic	140.00	95.00	250	35000.00	Cat B	Fast
19	Pantobert D Tab	Gastric Care	130.00	88.00	259	33670.00	Cat B	Fast
20	Dolo Pain Spray 50g	Pain Relief	195.00	140.00	168	32760.00	Cat B	Non-Moving
21	Intapept in Syrup 60ml	Pediatric Care	150.00	104.00	171	25630.00	Cat B	Non-Moving
22	Patobert 40 Tab	Gastric Care	110.00	75.00	233	25509.00	Cat B	Slow
23	Alamin Plus Cap	Supplement	130.00	89.00	170	22100.00	Cat C	Non-Moving
24	Moxeb Eye Drop	Eye Care	105.00	72.00	175	18450.00	Cat C	Non-Moving
25	Motinor LC Syrup 60ml	Pediatric Care	95.00	65.00	193	18335.00	Cat C	Slow
26	Clobeta GM 10g Cream	Dermatology	85.00	58.00	212	18020.00	Cat C	Slow
27	Moxib D Eye Drop	Eye Care	110.00	78.00	152	16720.00	Cat C	Non-Moving
28	Mahana c Tab	Pain Relief	72.00	48.00	222	15984.00	Cat C	Slow
29	Dexalife Inj 30ml	Injectable	73.00	48.00	208	15184.00	Cat C	Slow

30	Gentala b 30ml Inj	Injectab le	45.00	28.00	169	7605.00	Cat C	Non-M oving
----	--------------------------	----------------	-------	-------	-----	---------	-------	----------------

Table 5: Complete SKU Master Inventory Ledger — Anand Pharma (Jan–June 2026)